

Racing the Clock: Teaching VLNs Budget-Conditioned Object Search

Maitree Meher¹, Siming He¹, S. Shankar Sastry¹, Jitendra Malik¹, Claire Tomlin¹, Jonas Frey^{1,2}

Abstract—How do action budgets shape vision-language navigation (VLN) exploration, and how does the source of supervision affect learned search? We study *budget-conditioned object search*: finding as many distinct instances of a requested category as possible within a budget, with discovery defined geometrically by camera field of view and proximity. Distinct discoveries provide a denser evaluation signal than single-target success. We compare two supervision schemes: an *object-level Bellman expert* routes among targets with the full map and locations, while a *room-level Bellman expert* represents unexplored rooms through semantic estimates and cross-scene priors, leaving their unseen contents unknown. Both optimize budgeted Bellman objectives within their scene abstractions. We distill them into Uni-NaVid under matched training conditions and also evaluate greedy-frontier supervision on held-out HM3D scenes. Changing only the stated budget changes learned trajectories. Despite higher object-level expert return, room-level supervision achieves the highest mean collection among evaluated VLNs at every tested budget. At 2000 steps, it finds 72.7% of reachable targets, versus 68.5% under greedy-frontier supervision, 65.5% under object-level supervision, and 44.8% for the pretrained VLN. These results show that expert-return rankings need not predict supervision quality for VLN exploration and motivate accounting for budget and expert search behavior when collecting demonstrations.

I. INTRODUCTION

An indoor search robot must allocate its budget between nearby targets and distant regions: a short budget may favor immediate collection, while a longer one may justify travel to a promising room. To assess both effective budget use and its influence on decisions, we first ask: *how does the available budget change where and how a vision-language navigation (VLN) policy searches?*

This motivates *budget-conditioned object search*: finding as many distinct instances of a requested category as possible within an action budget. The agent receives RGB history, a target instruction, and a remaining-budget countdown, but no complete map or unseen target locations. Travel, scanning, and target approach consume budget. We count an instance as *collected* when it meets geometric proximity and camera-field-of-view criteria, without physical pickup. Distinct discoveries provide a denser signal than binary success and preserve repeated choices between visible targets and further exploration. Collection across budgets measures search performance; controlled countdown interventions test behavioral sensitivity (Fig. 1).

The remaining horizon can change the optimal action despite identical observation histories [1]; omitting it creates *state aliasing* between short-horizon collection and longer-horizon exploration. Although finite-resource planning is

well studied, pretrained VLNs such as Uni-NaVid [2] lack an explicit remaining-budget countdown. In a NaVILA [3] pilot, a shorter stated limit sometimes produced more actions than a longer one, motivating budget-aware demonstrations beyond merely adding a prompt limit.

To assess what supervision best transfers to a partially observed student, we next ask: *what kind of expert demonstrations effectively teach budget-conditioned search?* A full-map expert can route efficiently yet prescribe different actions in environments indistinguishable to the student, an information mismatch that can hinder asymmetric imitation [4], [5], [6], [7]. This motivates comparing demonstrations that route among known targets with demonstrations that search regions for undiscovered targets. The empirical question is how the supervision schemes transfer to the same VLN.

Equal target values and unit primitive-action costs isolate these questions from target priorities and cost weights, while collection and uncertain exploration still compete for finite budget. Our focus is supervision and budget use within a fixed video VLN architecture. Multi-room search preserves the key choice between a visible target and a room with uncertain, delayed reward. In our room-level abstraction, doorway evidence estimates room category, and cross-scene priors predict discoveries while unseen object contents remain hidden. This representation specifies both the scale of planning and the information available for choosing where to search.

Using the same finite-budget Bellman principle, an *object-level expert* routes among targets with the full map and locations, whereas a *room-level expert* selects regions from observed connectivity, doorway evidence, and cross-scene priors. Their Uni-NaVid students predict primitive actions from RGB history and language; a greedy frontier expert and its supervised VLN provide coverage-oriented comparisons.

All three expert-supervised VLNs use matched scenes, budgets, and record counts; the two Bellman students also use identical optimization. Room-level supervision achieves the highest mean collection at every budget in ten held-out HM3D scenes despite lower expert return. Budget-blind controls distinguish countdown gains from supervision-source differences; countdown interventions test behavioral sensitivity. These findings motivate recording demonstration budgets and considering expert search behavior when choosing supervision.

Our contributions are:

- We formulate *budget-conditioned object search*, using distinct target collection to measure effective budget use and controlled countdown interventions to assess

¹UC Berkeley

²Stanford University



Fig. 1. **The stated budget changes the learned VLN’s search strategy.** Room-level Bellman-supervised Uni-NaVid is rolled out from the same start pose in the same held-out HM3D scene with identical deterministic decoding; only the budget in the instruction changes. The resulting trajectories differ substantially rather than simply extending one another, most visibly between $B = 150$ and $B = 400$. Across $B = 50/150/400/800/2000$, the policy collects 4/11/10/12/12 of 13 reachable chairs.

behavioral sensitivity to the budget.

- We instantiate a room-level Bellman expert that plans with uncertain target yields using observed evidence and cross-scene priors, providing budget-conditioned demonstrations for a partially observed VLN.
- We compare room-level, fully informed object-level, and greedy-frontier supervision for Uni-NaVid on matched scenes, budgets, and record counts. Room-level supervision yields the highest mean collection, while higher expert return need not yield a stronger student.

II. RELATED WORK

A. Privileged and Planning Supervision

A high-performing expert need not provide optimal supervision to a student with partial information. ADVISOR [4] and A2D [5] reduce reliance on privileged actions or combine imitation with learning under partial observability; recent work studies student realizability and teacher–student alignment [6], [7]. Privileged planning also supervises navigation in Learning by Cheating [8], shortest-path imitation in SPOC [9], and large-scale exploration [10]. We study how planning-based supervision transfers to a fixed video VLN, comparing fully informed object-level routing with room-level search under uncertainty. The two Bellman-supervised students share architecture and training settings. Greedy-frontier supervision provides an additional comparison with coverage-oriented exploration demonstrations.

B. Vision-Language Navigation and Object Search

Large-scale imitation and reinforcement learning support policies such as ViNT and LM-Nav [11], [12], SPOC [9], and video VLN NaVid [13], Uni-NaVid [2], and NaVILA [3]. Object search exploits geometry and semantics [14], [15], with POMDPs modeling location uncertainty [16]. Exploration uses rooms and doorways [17] and hierarchical frontier planning [18]. MultiON [19], GOAT-Bench [20], and long-horizon VLN [21] extend task horizons. Whereas object search commonly tests single-target success under a fixed limit [22], we measure distinct discoveries and search-strategy changes across budgets. Our comparisons examine supervision for a video VLN; explicit planning methods supply demonstrations and reference trajectories.

C. Budgeted and Budget-Conditioned Exploration

Finite-resource planning includes orienteering [23], constrained MDPs [24], and informative paths trading informa-

tion gain against time, distance, or energy [25], [26], [27], [28]. Predicted maps and pathwise information gain support reasoning about unseen regions [29], alongside frontier-based methods [30], [31].

Decision Transformer [32] and reward-conditioned or constrained policies [33], [34] condition on returns, horizons, or constraints. Time Limits in RL [1] explains horizon-induced state aliasing; navigation methods incorporate budget, return, or termination signals [35], [36], [37], [38]. We focus on *exploration*: which regions to search and in what order as the budget changes, and which expert supervision transfers this behavior to a video VLN.

III. BUDGET-CONDITIONED OBJECT SEARCH

A. Problem formulation

An episode $e = (\mathcal{M}, q_0, y, B)$ specifies a partially observed multi-room environment, initial agent state, target category, and action budget. Rooms organize exploration; the objective counts object instances and the policy selects primitive actions. The budget can also represent time, distance, energy, or another resource.

At step t , the policy $\pi(a_t \mid h_t, y, b_t)$ selects $a_t \in \mathcal{A}$ from observation–action history $h_t = (o_0, a_0, \dots, o_t)$ and remaining budget $b_t = B - \sum_{k=0}^{t-1} c(a_k)$, with positive action cost $c(a_t)$. Here π denotes any policy using these inputs; our learned instances are Uni-NaVid students trained on expert actions (Sec. V-C). The full environment is unobserved, so different environments may be indistinguishable under the same history.

Let $\mathcal{I}_y(\mathcal{M})$ contain the category- y instances and $\mathcal{F}_t \subseteq \mathcal{I}_y(\mathcal{M})$ those found by time t under a policy-independent criterion. An instance is collected when it is first credited by this criterion (Sec. V-A). The reward $r_t = |\mathcal{F}_{t+1} \setminus \mathcal{F}_t|$ counts only newly credited instances after action a_t ; revisiting a credited instance adds no reward. The objective is

$$\pi_B^* \in \arg \max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^{\tau-1} r_t \right] \quad \text{s.t.} \quad \sum_{t=0}^{\tau-1} c(a_t) \leq B, \quad (1)$$

where τ denotes the termination time.

Tight budgets may favor immediate reward; longer budgets can justify exploration with delayed reward. Optimal actions therefore depend on both observations and remaining budget, not merely a trajectory’s truncation point.

Heterogeneous target values replace the count with $r_t = \sum_{i \in \mathcal{F}_{t+1} \setminus \mathcal{F}_t} w_i$, $w_i \geq 0$; Eq. (1) already permits nonuniform

positive costs. We use $w_i = c(a_t) = 1$. Extensions require corresponding expert reward/cost estimates and policy inputs conveying the task specification. They preserve our research questions, but persistence of the supervision advantage requires further evaluation.

B. Budget-conditioned evaluation

Counting additional discoveries distinguishes policies beyond single-target success and provides a denser evaluation of budget use.

For an episode e , let $\mathcal{R}_y(e)$ denote the set of target instances considered valid and reachable under the benchmark’s embodiment and discovery criterion. For a fixed evaluated policy, $\mathcal{F}_e(B)$ is the set of distinct instances credited by termination of the rollout with initial budget B , i.e., $\mathcal{F}_t$ at that rollout’s termination time. Changing B can change both the trajectory and its termination time, hence the discovered set. We define normalized collection, the fraction of valid reachable instances found, as

$$N_e(B) = \frac{|\mathcal{F}_e(B) \cap \mathcal{R}_y(e)|}{|\mathcal{R}_y(e)|}, \quad (2)$$

and report its average over the set $\mathcal{E}_B$ of matched evaluation episodes run with budget B ,

$$N(B) = \frac{1}{|\mathcal{E}_B|} \sum_{e \in \mathcal{E}_B} N_e(B). \quad (3)$$

Higher collection with more budget does not establish use of the budget input: a fixed strategy may find more objects simply by running longer. We therefore hold scene, target, start, model, and decoding fixed, vary only the countdown, and examine trajectory divergence within the shared horizon. These interventions test behavioral sensitivity; $N(B)$ measures performance.

IV. ROOM-LEVEL BELLMAN EXPERT

To generate search demonstrations, we instantiate an expert that retains spatial and semantic structure for planning while discovering unseen targets through budget-dependent exploration.

The abstraction $z = \varphi(\mathcal{M})$ specifies retained structure and hidden details. The expert’s information state s_t summarizes what has been revealed about z to the expert. Multiple environments can remain compatible with s_t , giving expected rewards and transitions

$$R_\varphi(s, a) = \mathbb{E}[r_t \mid s_t = s, a_t = a], \quad (4)$$

$$P_\varphi(s' \mid s, a) = \Pr(s_{t+1} = s' \mid s_t = s, a_t = a). \quad (5)$$

For remaining budget b , the corresponding Bellman recursion is

$$Q_\varphi^*(s, b, a) = R_\varphi(s, a) + \mathbb{E}_{s' \sim P_\varphi(\cdot \mid s, a)} [V_\varphi^*(s', b - c(a))], \quad (6)$$

$$V_\varphi^*(s, b) = \max_{a: c(a) \leq b} Q_\varphi^*(s, b, a), \quad V_\varphi^*(s, 0) = 0. \quad (7)$$

Shared planning principle across scene abstractions. Both Bellman experts optimize immediate collection plus

continuation value under a finite budget. The object-level expert routes among known targets using the full map and true locations. The room-level expert represents unexplored rooms through observed connectivity and doorway-based category estimates. Cross-scene priors predict discoveries, execution costs, and successor states while unseen objects and locations remain unknown. This abstraction specifies both planning units and information access. The room-level implementation estimates Bellman values from cross-scene transitions, as detailed below; it does not claim an exact solution of the original partially observed environment.

Each expert supplies action labels to train a separate UniNaVid student with the same imitation objective. At inference, students act from RGB history and language, including the countdown, without querying an expert or running its planner. The room-level expert additionally maintains an explicit room graph and semantic state; expert and student inputs therefore differ.

A. Room-Level Instantiation

Room entry incurs travel and search costs, reveals targets, and may expose further regions; semantics provide target-yield priors. This captures the collection–exploration tradeoff without requiring a room representation in the task objective or student’s action space. Figure 2 shows entered rooms and adjacent unentered candidates, evaluated by expected yield and the budget needed to reach and search them.

Our room-level φ retains observed connectivity and semantics while withholding the unrevealed geometry, contents, and target locations of unexplored rooms. The information state s_t contains the observed room graph $G_t^{\text{obs}} = (V_t^{\text{obs}}, E_t^{\text{obs}})$, robot state, observed room types and discovered contents, and frontier-doorway evidence. Observed room types and objects come from simulator ground-truth annotations, exposed as rooms are entered or objects become visible, rather than inferred from RGB. Entering a room does not reveal its complete object inventory; only visible objects become available for direct approach.

The frontier set $\mathcal{U}(s_t)$ contains adjacent unentered rooms on the current floor with a doorway reachable along a finite geodesic. An unentered room may have objects visible through its doorway; this limited evidence does not reveal its complete contents. Such objects provide a room-type estimate through the cross-scene prior $P(\rho \mid o)$ learned from Matterport3D [39], where ρ is room type and o is doorway-visible object evidence. The expert uses this estimate to predict discoveries in the unobserved interior. An estimated full sweep exceeding the budget does not exclude a frontier; the budget instead filters transition outcomes below.

B. Cross-Scene Bellman Estimation

We estimate the unknown reward and transition models from a bank of room-transition records from 181 HM3D scenes. Each record $\xi_i = (s_i, v_i, r_i, d_i, s'_i)$ stores the information state before selecting frontier room v_i , the newly credited target counts r_i , the realized primitive-action cost d_i , and the information state after execution. Thus, r_i^y is

a) Only entered rooms are known

b) Each room frontier door is priced

c) Sweeps are interrupted, and end early

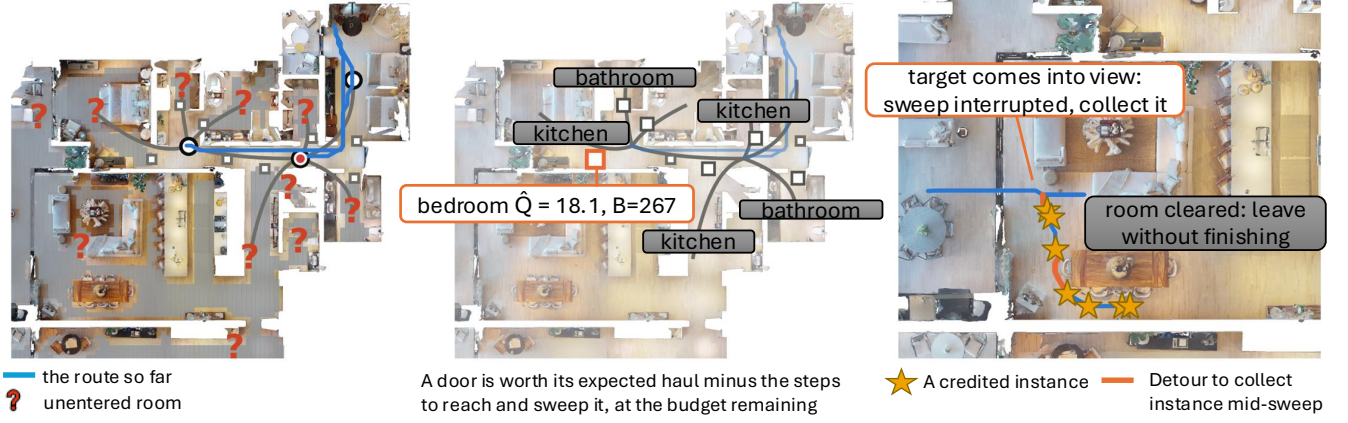


Fig. 2. **Room-level Bellman expert.** (a) “Known” rooms have been entered; their recorded contents include only discovered objects. (b) Frontier rooms are valued using doorway-visible semantic evidence and cross-scene Bellman estimates. (c) Execution remains closed-loop: visible targets can interrupt a sweep, after which the expert replans.

a room-transition reward for category y , accumulating the primitive-step rewards defined in Sec. III. These outcomes supply empirical samples for Eqs. (6)–(7); the doorway prior supplies the candidate’s estimated room type, not its realized reward. This bank supports expert planning; the video/action records used to train the VLN are described in Sec. V-C.

For a query (s, v) , we retrieve transitions with matching abstract context–action signatures and use weighted nearest neighbors when exact matches are sparse. Retrieval is strictly cross-scene. The source scenes are split into five folds, and planning in a scene excludes every transition from that scene’s entire fold. All benchmark scenes are excluded from the evaluation transition bank.

Retrieved transitions jointly sample reward, execution cost, and successor context. Let $\mathcal{N}_b(s, v)$ contain those with $d_i \leq b$, with weights $\alpha_i \geq 0$ normalized to $\sum_{i \in \mathcal{N}_b(s, v)} \alpha_i = 1$. For category y , the empirical backup combines immediate reward and continuation return:

$$\hat{Q}_y(s, b, v) = \sum_{i \in \mathcal{N}_b(s, v)} \alpha_i \left[r_i^y + \hat{V}_y(s'_i, b - d_i) \right], \quad (8)$$

where r_i^y is the immediate room-transition reward for category y and $\hat{V}_y(s'_i, b - d_i)$ estimates the best subsequent return after spending d_i steps. The state value selects the highest estimated return over frontier rooms and a zero-valued stopping option:

$$\hat{V}_y(s, b) = \max \left(0, \max_{v \in \mathcal{U}(s)} \hat{Q}_y(s, b, v) \right), \quad \hat{V}_y(s, 0) = 0. \quad (9)$$

The hats denote estimates computed recursively under the empirical model. Filtering by $d_i \leq b$ and renormalizing conditions on outcomes that fit the budget; it does not model the probability or reward of exceeding it.

A room with no admissible transition has undefined $\hat{Q}_y$ and is excluded from the maximization, but remains eligible for the fallback below.

Continuation enables non-myopic exploration: a low-reward connector room may expose promising regions, yet

be unhelpful under a tight budget. Positive transition costs permit evaluation in increasing order of b .

Doorway evidence thus estimates room type; retrieved outcomes turn this semantic expectation into reward, cost, and continuation estimates. A room’s ranking depends on budget-dependent returns, not its semantic promise alone.

C. Closed-Loop Execution

Primitive execution is closed-loop: a newly visible uncredited target within 5 m and line of sight interrupts the room sweep for approach-and-collect. The executor uses observed targets and coverage to determine when its room search is complete; it does not consult a ground-truth room inventory or target count. Figure 2(c) shows a mid-sweep detour and departure with most sweep viewpoints unvisited. Alternating scan directions at successive viewpoints reduces the left:right turning ratio from 2:1 to 1.2:1.

If no frontier has positive estimated value or admissible evidence, the expert enters the nearest unexplored frontier as a curiosity fallback. The zero branch in Eq. (9) is a value-estimation option, not an execution rule requiring termination when positive evidence is absent. After every room transition or interruption it updates s_t and replans. If all frontiers are exhausted while budget remains, it re-sweeps visited rooms, providing demonstrations at horizons longer than required to explore the scene.

Unexplored contents and target locations are used only after execution to construct transitions, never for action selection. The abstraction merges distinct histories and the model uses finite cross-scene data, so this is an *empirical room-level Bellman expert*, without a Bayes-optimality claim for the original partially observed environment.

V. EXPERIMENTS

We test budget-dependent VLN behavior through collection scores and countdown interventions, and supervision transfer through matched training comparisons. Experts and learned policies share the objective, budgets, and discovery

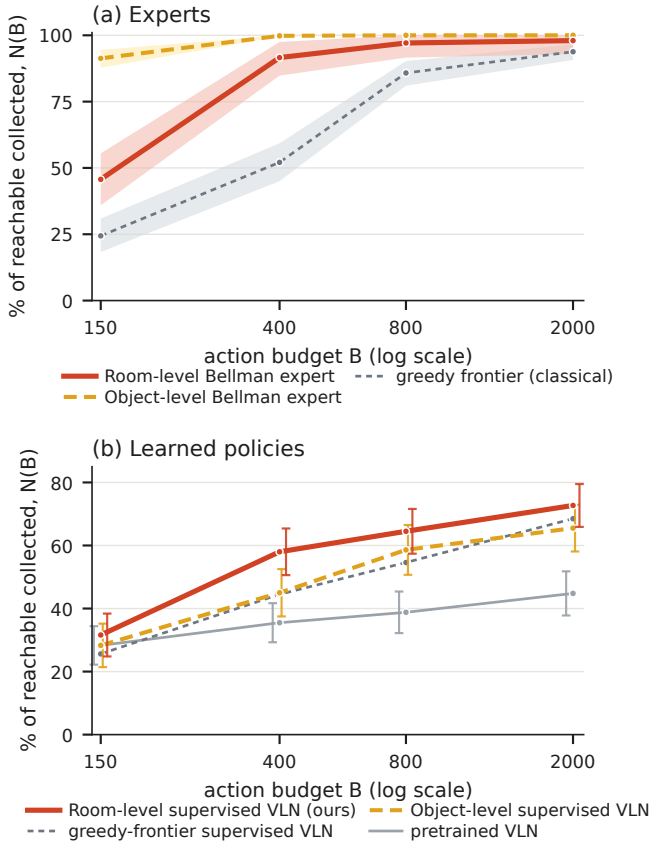


Fig. 3. **Expert return and supervision transfer on the frozen benchmark.** (a) Expert collection $N(B)$ with 95% bootstrap intervals: object-level Bellman leads room-level Bellman, which leads unprivileged frontier baselines. (b) Mean collection after matched Uni-NaVid fine-tuning reverses the ordering of the two Bellman experts. Greedy frontier is a non-learned reference; Table I additionally reports greedy-frontier-supervised VLN performance. Budget-dependent margins and long-horizon revisiting are discussed in Sec. V-C.

criterion. The two Bellman experts also share the budgeted optimization principle, applied to known targets or expected room yields according to their available information (Sec. IV); greedy frontier provides a coverage-oriented baseline. We examine demonstrations before learned policies, using the held-out HM3D protocol below.

A. HM3D Experimental Setup

Environment and action space. We instantiate budget-conditioned object search in multi-room HM3D [40] indoor scenes with $\mathcal{A} = \text{FORWARD, LEFT, RIGHT, STOP}$, where FORWARD moves 0.25 m and LEFT/RIGHT rotate 30° . Each primitive costs one budget unit, and episodes terminate on STOP or budget exhaustion. We evaluate $B \in \{150, 400, 800, 2000\}$. Learned VLN receive the remaining budget only through language, e.g., “Find as many chairs as possible. You have b_t steps remaining,” with no dedicated numerical budget input. Their observations consist of RGB video history; the full map, room graph, and ground-truth locations of unseen targets are not provided to them.

Discovery criterion. We define an object as found using a policy-independent geometric referee. A target instance is credited when the agent is within 1.5 m of its bounding

extent, within 1.5 m vertically, and its bearing is within 39.5° of the camera optical axis. This bound is half of the camera’s 79° horizontal field of view. The criterion measures target coverage within the camera’s nearby viewing region. It is applied identically to every policy and requires no separate recognition report or collection action. Each instance is credited at most once. We exclude instances in the starting room and those without any valid crediting pose geodesically reachable from the start. Reachability is computed directly on the HM3D navmesh rather than the room graph, allowing, e.g., an instance visible through a doorway to be credited without entering its semantic room.

Matched evaluation protocol. We evaluate all methods on 10 held-out HM3D-val scenes excluded from both VLN fine-tuning and room-level Bellman transition data. The benchmark contains 50 admission-verified (scene, category) pairs with two deterministic starts each. Scene, category, initial position, heading, and budget are matched across methods, giving 100 episodes per budget. Table I reports mean normalized collection $N(B)$ (Sec. III) with the half-width of a 95% confidence interval (CI). CIs are percentile bootstraps over evaluation episodes: for each method and budget we resample the matched episode set 10,000 times with replacement and take the 2.5th and 97.5th percentiles of the resampled mean. The episode is the resampling unit, so scene, category, and start are resampled together. Fig. 3(a) draws these intervals as bands and Fig. 3(b) as error bars, offset horizontally for legibility. We also examine cross-budget trajectory divergence, revisiting, and monotonicity violations, where a larger budget yields fewer targets.

B. Expert Supervision

Expert performance. The object-level Bellman expert solves budgeted routing with the full map and target locations; the room-level expert follows the information constraints in Sec. IV. The greedy frontier expert provides a coverage-oriented exploration baseline with explicit spatial memory. We evaluate both its direct execution and the VLN trained on its demonstrations under the same held-out protocol and action budgets.

The object-level expert leads at every budget, most strongly at tight budgets (Fig. 3(a)). Both Bellman experts act until the budget is exhausted: after collecting all targets, the object-level expert re-sweeps the environment, while the room-level expert re-sweeps visited rooms after exhausting frontiers. These continuation behaviors are part of the respective supervision schemes. Re-sweeping ensures every demonstration reaches its assigned budget, controlling the distribution of remaining budgets in training records across experts despite different completion times.

Budget-dependent demonstrations. Varying only B at matched scenes, target categories, and starts, 55% of 691 admitted training pairs select a different first room across budgets; 96% are monotone in targets collected.

At equal executed-action counts, mean cumulative collection for $B = 150$ leads larger-budget demonstrations for about 131 steps, by up to 7.0 points, before saturating. This

controls for additional execution time: tight budgets favor immediate yield, while larger ones justify delayed-reward transfers. Thus, the demonstrations contain budget-dependent decisions rather than truncations of a common route.

C. Learning a Budget-Conditioned VLN

Fine-tuning setup. Bellman-expert demonstrations are converted to Uni-NaVid’s native training format using its fixed inference cadence: after every two executed primitives, the model is queried for the next four actions. Each record contains the video history up to time t , the expert’s next four primitives, and a natural-language countdown with $B - t$ steps remaining. The budget is provided only through language; we introduce no numerical budget input or architectural modification.

The room-level Bellman supervision has 388,199 records: 214,562 from 89 HM3D and 173,637 from 59 Matterport3D scenes, at budgets 50, 150, 400, 800, 1200, and 2000. A deterministic 20% of records omit the countdown. We train directly on this mixture without contrast mining or reweighting. For each Bellman supervision scheme, we also train a *budget-blind* control using the same video/action records, with the countdown removed from every record. Architecture, optimization, and training steps are fixed; only the training countdown text differs. The controls retain budget-dependent trajectories but receive no explicit countdown during training.

We otherwise follow the Uni-NaVid stage-2 recipe: the vision encoder is frozen, the language model and projector are trained, the learning rate is 10^{-5} , and training lasts one epoch, 3,032 steps at effective batch size 128, taking approximately 62 hours on four RTX PRO 6000 Blackwell GPUs. The object-level, room-level, and greedy-frontier students use the same scenes, budgets, and record count. The two Bellman students also share architecture and optimization; only their expert supervision differs.

Learned-policy performance. Table I compares experts and learned policies. The non-learned greedy frontier uses explicit spatial memory, whereas VLNs retain history implicitly in context. Its direct execution provides a long-horizon coverage reference; its supervised VLN tests transfer of these demonstrations to RGB history. The central comparison concerns supervision within the Uni-NaVid architecture.

At $B = 150/400/800/2000$, room-level supervision improves over the pretrained VLN by 3.3/22.5/25.7/27.9 points, over object-level supervision by 3.3/13.0/5.9/7.2, and over greedy-frontier supervision by 6.0/13.5/9.9/4.2; its margin over the pretrained VLN widens with budget. At $B = 2000$, it yields 72.7% collection, versus 68.5% for greedy-frontier supervision, 65.5% for object-level supervision, and 44.8% for the pretrained VLN. All VLN beat the non-learned greedy frontier expert at $B = 150$, only room-level supervision does at $B = 400$, and the frontier expert leads every VLN from $B = 800$ onward (Table I).

The ordering of the two Bellman experts therefore reverses after distillation: higher object-level expert return is associated with lower student collection, while room-level

TABLE I
BUDGET-CONDITIONED OBJECT-SEARCH PERFORMANCE (MEAN $\pm$ 95% CI HALF-WIDTH): DEMONSTRATION EXPERTS AND UNI-NAVID SUPERVISION VARIANTS.

Policy	$B = 150$	$B = 400$	$B = 800$	$B = 2000$
(a) Expert policies				
Greedy frontier expert	24.4 $\pm$ 6.3	52.1 $\pm$ 7.1	85.8 $\pm$ 4.6	93.8 $\pm$ 2.9
Object-level Bellman expert	91.3 $\pm$ 3.4	99.8 $\pm$ 0.2	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0
Room-level Bellman expert (ours)	45.7 $\pm$ 9.8	91.6 $\pm$ 6.4	97.1 $\pm$ 4.2	98.0 $\pm$ 3.5
(b) Learned VLN policies				
Pretrained VLN	28.3 $\pm$ 6.1	35.5 $\pm$ 6.2	38.8 $\pm$ 6.6	44.8 $\pm$ 7.0
VLN trained on greedy frontier expert	25.6 $\pm$ 6.0	44.5 $\pm$ 7.3	54.6 $\pm$ 7.5	68.5 $\pm$ 7.3
VLN trained on object-level expert	28.3 $\pm$ 6.2	45.0 $\pm$ 7.3	58.6 $\pm$ 7.5	65.5 $\pm$ 7.4
budget-blind training	25.5 $\pm$ 6.0	42.2 $\pm$ 7.2	55.0 $\pm$ 7.5	61.8 $\pm$ 7.5
VLN trained on room-level expert (ours)	31.6 $\pm$ 6.8	58.0 $\pm$ 7.4	64.5 $\pm$ 7.1	72.7 $\pm$ 6.8
budget-blind training	28.5 $\pm$ 6.2	51.6 $\pm$ 7.5	59.5 $\pm$ 7.5	63.5 $\pm$ 7.5

Panel (a): experts. Panel (b): pretrained Uni-NaVid or fine-tuned on the named expert; indented rows repeat that supervision with the countdown removed from every training record. Values are mean $N(B)$ (%) $\pm$ the half-width of the 95% confidence interval. $N(B)$ is the percentage of distinct valid, reachable targets found within B primitives, averaged across the 100 matched episodes. Higher is better; bold marks the highest mean among evaluated VLN at each budget.

performance exceeds greedy frontier before and after distillation. At $B = 2000$, the greedy-frontier-supervised VLN also beats the object-level student (68.5% versus 65.5%), further showing that expert-return rankings need not transfer; this comparison does not isolate Bellman planning from other demonstration differences.

The comparison evaluates object-level routing and room-level search as complete supervision schemes, including their abstractions and execution behaviors. Information alignment is one possible explanation for the reversal [4], [5]: room-level demonstrations search for unseen targets using semantic expectations, whereas object-level routes use their known locations. The empirical result establishes that expert-return rankings need not transfer; it does not attribute the difference to information access alone.

Budget-blind training. Within each expert’s fixed demonstrations, countdown annotations improve mean collection at $B = 150/400/800/2000$ by 2.8/2.8/3.6/3.7 percentage points for object-level and 3.1/6.4/5.0/9.2 for room-level supervision (Table I). With countdowns omitted from both students’ training, room-level supervision still leads by 3.0/9.4/4.5/1.7 points. Thus, its advantage persists without explicit budget annotations, while countdown gains depend on supervision source. These comparisons distinguish supervision-source and countdown effects while retaining budget-dependent expert trajectories. The next intervention tests countdown use at inference.

Controlled countdown intervention. To test use of the budget input independently of collection scores, matched probes fix the initial observation, model, and deterministic decoding while changing only the countdown sentence.

Changing $B = 150$ to $B = 2000$ changes the room-level Bellman-supervised VLN’s trajectory on all six held-out-scene pairs, with divergence after 2–11 actions. Collection increases on every pair: $2 \rightarrow 8$, $4 \rightarrow 7$, $1 \rightarrow 10$, $3 \rightarrow 12$, $1 \rightarrow 7$, and $4 \rightarrow 11$. Same-budget repeats reproduce the action sequence. Divergence before either budget expires establishes countdown sensitivity; collection gains also reflect additional available actions and do not alone establish more

TABLE II
REAL-ROBOT CHAIR SEARCH OVER FIVE RUNS PER BUDGET.

Budget B	50	150	400	2000
Chairs found	3.4 ± 1.14	5.0 ± 1.00	7.6 ± 1.52	9.2 ± 1.30
Mean SR (%)	24.3	35.7	54.3	65.7

efficient allocation. In the mesh-rendered probe of Fig. 1, the $B = 50/150/800$ routes diverge within the first few actions from the same start and collect 4/11/12 of 13 chairs.

Across 206 mined contrast pairs, balanced first-action accuracy against the room-level expert’s budget-conditioned actions rises from 0.344 pretrained to 0.572 after room-level supervision. Yet on the 84 pairs where the expert changes its first action, correctly predicting both branches remains near chance. Budget conditioning is thus more reliable over the evolving closed-loop trajectory than as an immediate switch at $t = 0$.

Failure analysis. A substantial expert–student gap remains, especially at long horizons. At $B = 800$ and $B = 2000$, 92–93% of the VLN’s forward actions revisit traversed cells. Offline imitation trains on expert trajectories, providing limited coverage of states reached after student deviations. This pattern motivates better closed-loop recovery during learned search.

D. Real-World Deployment

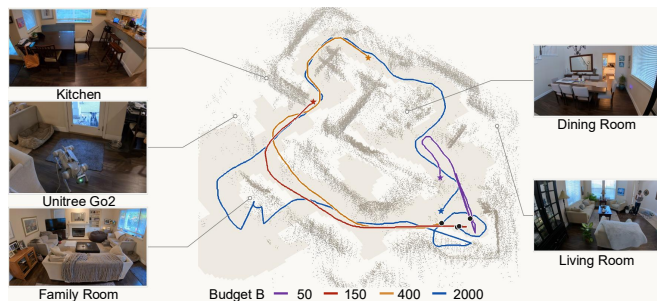


Fig. 4. **Real-robot deployment in a multi-room home.** Routes at $B \in \{50, 150, 400, 2000\}$ share the scene, start, target category, and weights. At $B = 50$, the robot searches for the closest chairs near its start; with larger budgets, it first makes longer excursions toward the family room and kitchen. Routes overlay the robot’s lidar map; circles mark starts, stars mark endpoints, and photographs show traversed spaces.

We deploy the room-level Bellman-supervised VLN zero-shot on an indoor quadruped without real-world fine-tuning. Following Uni-NaVid [2], an onboard camera streams RGB to a remote GPU server, which returns four primitives per call in 0.22 s on average; locomotion dominates runtime.

As in simulation, forward and turning actions command 0.25 m translation and 30° rotation. Odometry executes each primitive until its measured displacement reaches the target, maintaining consistency with predictions.

In a multi-room home containing 14 chairs, we fix the start pose, target, and weights and vary only $B \in \{50, 150, 400, 2000\}$ (Fig. 4). We run each budget five times; Table II reports the mean and standard deviation (SD) of chairs found. We define the success rate in each run as $SR = 100 \times (\text{chairs found}/14)$ and report its mean.

Mean discoveries rise from 3.4 to 9.2 and SR from 24.3% to 65.7% as the budget increases. Routes also differ: at $B = 50$, the robot searches near its start in the living and dining rooms, whereas larger budgets permit longer transfers to the family room and kitchen. The repeated trials and route changes show budget-dependent behavior despite physical-camera observations and locomotion error.

VI. CONCLUSION

We study budget-conditioned VLN exploration and the transfer of different expert supervision schemes to Uni-NaVid. Budget-conditioned object search counts distinct geometric discoveries for a denser evaluation signal than binary success; controlled countdown interventions show that changing only the stated budget changes trajectories, beyond extending a fixed route.

Expert return and supervision quality diverge: despite higher return from the fully informed object-level Bellman expert, room-level supervision achieves the highest mean collection among evaluated VLNs at every budget. Greedy-frontier supervision also beats object-level supervision at the longest budget, reinforcing that expert rankings need not transfer. The matched Bellman comparison supports room-level search as an effective demonstration scheme for the evaluated VLN; information access, abstraction, and execution jointly define that scheme. Budget-blind controls show that room-level supervision retains a mean advantage without countdown annotations during training, while including these annotations improves both students at every tested budget.

For real-world demonstrations, these findings motivate recording budget with observations and actions and considering search under uncertainty. Such metadata both explains choices between nearby targets and other rooms and provides a learning signal; whether it improves real-world learning requires further evaluation.

At 2000 steps, the room-level-supervised VLN remains below its expert (72.7% versus 98.0%). Frequent revisiting and the non-learned frontier expert’s long-horizon advantage motivate better spatial memory and recovery from trajectory deviations. Rooms are an empirical abstraction, and evidence is limited to indoor search with equal-value targets and action-count budgets. Weighted target rewards and nonuniform action costs extend the formulation, but supervision rankings there remain unknown. Future work can evaluate these settings and select abstractions for downstream student performance.

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